

Roman Yoder

 rayproengineering.com  rayproengineering@gmail.com  (480) 249-2201

Professional Summary

Mechanical engineer with professional software engineering experience in robotics and automation. Skilled in control systems, path planning, and industrial/medical applications. Passionate about scalable solutions for complex interdisciplinary problems. Strong communicator and collaborator who thrives in multi-functional teams and fast-paced environments.

Education

M.S. Mechanical Engineering (Robotics)

GPA: 3.72

Texas A&M University

Aug 2021 – Dec 2023

B.S. Mechanical Engineering

GPA: 3.65

Colorado State University

Aug 2017 – May 2021

Experience

Medtronic – Software Controls Engineer, Surgical Robotics

Apr 2024 – Present

- Developed core control software for a tendon-driven flexible catheter used in robotic bronchoscopy.
- Contributed to a foundational understanding of catheter mechanics/kinematics.
- Transitioned early-stage research into robust, validated software as part of the New Product Development lifecycle.
- Filed multiple IP disclosures, including auto-homing, redundant DOF insertion control, buckling detection/prevention, and continuum force-sensing methods.
- Implemented delay pre-compensation algorithms to correct transmission latency and improve distal tip accuracy.
- Built a proof-of-concept safety system using ideal vs. measured insertion lengths inside a lung digital twin to detect obstruction risk.
- Contributed to the design of a modular Jacobian-based optimization pipeline enabling scalable, centralized control of robotic systems.
- Integrated a distributed fail-safe mechanism across multiple compute nodes for consistent system-level fault recovery.
- Mentored engineers and contributed technical guidance on emerging continuum-robotics initiatives.

Medtronic – Research Engineer

Spring 2023

- Designed and validated algorithms to generate efficient paths through lung vascular models.
 - Integrated these planners into an autonomous robotic catheter navigation prototype.
-

Research & Projects

Automated Welding Thesis

- Developed ROS-based welding automation pipeline using a 6-DOF Universal Robot and MIG welding tool.
- Created a custom Traveling Salesman Problem solver using Mixed Integer Programming to optimize weld sequences.
- Enabled human-robot collaboration by integrating user-guided part localization and adaptive trajectory execution.
- Ran simulations to test weld paths before deployment; validated results with physical prototypes.

Capstone – Jet Impingement Server Cooling

- Designed and built a prototype system to evaluate novel CPU cooling via jet impingement.
- Modeled and analyzed pressure drop and thermal performance using experimental and analytical methods.

Course Projects (Selected)

- **Reinforcement Learning:** Implemented actor-critic model to simulate real-time missile interception control.
 - **Non-Linear Controls:** Applied feedback linearization to stabilize high-speed missile trajectory.
 - **Robotic Manipulator Mechanics:** Solved multi-path routing using Dubbin's curves and TSP solver for ground vehicle.
-

Skills & Coursework

- **Robotics:** Kinematics, dynamics, path planning, RL, MDPs
- Mechatronic System Modeling
- Control System Design
- Optimization (Linear/Nonlinear)
- Vehicle Dynamics
- Combustion & Turbo Machinery
- **Tools:** Python, ROS, C++/OOP, **bash**, Simulink, Matlab
- Git, Copilot